

Reconstructing the canonical extension in the stable setting

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Joint work with Sam van Gool

The points of canonical extensions of doctrines, [arXiv:2606.28736](https://arxiv.org/abs/2606.28736)

Introduction

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Can infinite conjunctions be freely added to the syntax of a coherent theory?

And how does this affect the semantics of the given theory?

Canonical extensions of lattices

Canonical extensions provide an algebraic perspective on Stone and Priestley duality.

Given a Boolean algebra $\mathbb{B}$, the operation $\mathbb{B} \mapsto \mathbb{B}^\delta := \mathcal{P}(\mathbf{Ult}(\mathbb{B}))$ defines a left adjoint:

$$\mathbf{BA}^+ \begin{array}{c} \xleftarrow{\quad \quad} \\ \perp \\ \xrightarrow{\quad \quad} \end{array} \mathbf{BA}$$

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A *prime filter* of a lattice $F \subseteq L$ is an upwards closed subset such that:

$$\top \in F, \quad a, b \in F \implies a \wedge b \in F, \quad a \vee b \in F \implies a \in F \text{ or } b \in F.$$

Given a distributive lattice $\mathbb{D}$, the operation $\mathbb{D} \mapsto \mathbb{D}^\delta := \mathbf{Up}(\mathbf{PFilt}(\mathbb{D}))$ defines a left adjoint:

$$\mathbf{DL}^+ \begin{array}{c} \xleftarrow{\quad \perp \quad} \\ \xrightarrow{\quad} \end{array} \mathbf{DL}$$

Here, $\mathbf{DL}^+$ is complete lattices join-generated by completely join-prime elements, and complete morphisms.

N.B. $\mathbb{D}^\delta$ is join-generated by the principal up-sets of prime filters.

Coherent doctrines

Coherent doctrines are a predicate extension of Lindebaum-Tarski algebras.

Definition

A *coherent doctrine* is a functor $D: \mathcal{C}^{\text{op}} \rightarrow \mathbf{DL}$ such that:

- i. $\mathcal{C}$ has finite limits,
- ii. For each arrow $D(f)$ has a left adjoint $\exists_f$ satisfying:
 - a (Frobenius) $\exists_f(D(f)(a) \wedge b) = a \wedge \exists_f b$,
 - b (Beck-Chevalley) the square

$$\begin{array}{ccc} D(c \times_e d) & \xrightarrow{\exists_{\pi_d}} & D(d) \\ D(\pi_c) \uparrow & & \uparrow D(f) \\ D(c) & \xrightarrow{\exists_g} & D(c) \end{array}$$

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Examples

- i. If $\mathcal{C}$ is a (well-powered) coherent category, then $\text{Sub}_{\mathcal{C}}: \mathcal{C}^{\text{op}} \rightarrow \mathbf{DL}$ is a coherent doctrine.
- ii. Each coherent theory $\mathbb{T}$ induces a *syntactic doctrine* $D_{\mathbb{T}}: \mathbf{FinSet} \rightarrow \mathbf{DL}$ where $D_{\mathbb{T}}(X)$ is formulae in context X , up to $\mathbb{T}$ -equivalence.

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Definition

A *morphism of coherent doctrines* $(M, \alpha): D \rightarrow Q$ consists of:

- i. a lex-functor $M: \mathcal{C} \rightarrow \mathcal{C}'$;
- ii. and a natural transformation

$$\alpha: D \Rightarrow Q \circ M^{\text{op}}$$

such that the square

$$\begin{array}{ccc} D(c) & \xrightarrow{\exists_f} & D(d) \\ \alpha_c \downarrow & & \downarrow \alpha_d \\ QM(c) & \xrightarrow{\exists_{Mf}} & QM(d) \end{array}$$

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Example

The morphisms $D_{\mathbb{T}} \rightarrow \mathcal{P} := \mathbf{Sub}_{\mathbf{Set}}$ are just models of $\mathbb{T}$.

Canonical extensions of coherent doctrines

Definition (Coumans)

Write D^δ for the canonical extension of a doctrine, i.e. the functor

$$\mathcal{C}^{\text{op}} \xrightarrow{D} \mathbf{DL} \xrightarrow{(-)^\delta} \mathbf{DL}^+.$$

Proposition (Coumans)

The functor $D^\delta: \mathcal{C}^{\text{op}} \rightarrow \mathbf{DL}$ is a coherent doctrine, and $(\mathbf{id}_{\mathcal{C}}, \eta): D \rightarrow D^\delta$ is a morphism of coherent doctrines!

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We say $Q: \mathcal{C}'^{\text{op}} \rightarrow \mathbf{DL}$ is a DL^+ -doctrine if Q factors through $\mathbf{DL}^+ \subseteq \mathbf{DL}$.

If Q is DL^+ , every morphism of doctrines $(M, \alpha): D \rightarrow Q$ yields a natural transformation $\alpha^\sharp: D^\delta \Rightarrow Q \circ M^{\text{op}}$.

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If Q is DL^+ , every morphism of doctrines $(M, \alpha): D \rightarrow Q$ yields a natural transformation $\alpha^\sharp: D^\delta \Rightarrow Q \circ M^{\text{op}}$.

But: $(M, \alpha^\sharp): D^\delta \rightarrow Q$ is not necessarily a morphism of doctrines.

We do **not** obtain a left adjoint to $\mathbf{CohDoc}^+ \hookrightarrow \mathbf{CohDoc}$.

Example

The transformation $\mathbf{id}_{\mathcal{P}}^\sharp: \mathcal{P}^\delta \rightarrow \mathcal{P}$ is not a morphism of doctrines.

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commutes.

(These are Makkai's *p-models*,
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For any arrow f , we have $\exists_f = \exists_{G_f} \exists_{\pi_d}$ where G_f is the *graph* of f , i.e.

$$\begin{array}{ccccc} & & c & & \\ & \swarrow f & \downarrow G_f & \searrow & \\ d & \xleftarrow{\pi_d} & d \times c & \xrightarrow{\pi_c} & c, \end{array}$$

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Theorem

The morphism $(M, \alpha): D \rightarrow Q$ is extensible if and only if for every prime filter $\tau \subseteq D(c \times d)$, we have $\alpha_c^\sharp(\exists_{\pi_d} \tau) \leq \exists_{M\pi_d}(\alpha_{c \times d}^\sharp(\tau))$.

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Extensible points and ω -saturation

Let us specialise to a morphism $(M, \alpha): D_{\mathbb{T}} \rightarrow \mathcal{P}$, i.e. a model M of $\mathbb{T}$.

Here, a prime filter of $D_{\mathbb{T}}(X)$ is commonly called an X -type; e.g. $m \in M^X$, then $\text{tp}_M(m) = \{\varphi \mid M \models \varphi(m)\}$ is a prime filter.

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For finite sets X, Y and an $(X + Y)$ -type $\tau \subseteq D_{\mathbb{T}}(X + Y)$,

$$\alpha_X^\#(\exists_{\pi_Y}(\tau)) = \{m \in M^X \mid \text{for all } \varphi \in \tau, M \models \exists y. \varphi(m, y)\},$$

$$\exists_{M\pi_Y}(\alpha_{X+Y}^\#(\tau)) = \{m \in M^X \mid \text{there exists } n \in M^Y \text{ s.t. for all } \varphi \in \tau, M \models \varphi(m, n)\}.$$

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That is, $\tau(m, Y) := \{\varphi(m, y) \mid \varphi \in \tau\}$ is a Y -type *with parameters* $m \in M^X$,

- i. and $m \in \alpha_X^\#(\exists_{\pi_Y}(\tau))$ if $\tau(m, Y)$ is *consistent* with M ,
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Theorem

A model $(M, \alpha): D_{\mathbb{T}} \rightarrow \mathcal{P}$ is extensible if and only if every type with finitely many parameters that is consistent with M is realised in M , i.e. M is ω -saturated.

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Example

If $D_{\mathbb{T}}: \mathcal{C}^{\text{op}} \rightarrow \mathbf{DL}$ factors through *finite* distributive lattices, then $D_{\mathbb{T}} \cong D_{\mathbb{T}}^\delta$ and every point is extensible (cf. $M \models \mathbb{T}$ is ω -categorical $\implies M \models \mathbb{T}$ is ω -saturated).

Presheaf valued points

In classical model theory, we can also define X -types (i.e. prime filters) as *orbits* of elements $m \in \mathbb{M}^X$ under $\mathbf{Aut}(\mathbb{M})$, where $\mathbb{M}$ is a *monster model*.

Can we do the same for the canonical extension of a coherent doctrine?

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Just like Coumans, we will use *presheaf valued points*, i.e. morphisms $D_{\mathbb{T}} \rightarrow \mathbf{Sub}_{\mathbf{Set}}^{\mathcal{K}}$.

A morphism $(M, \alpha): D_{\mathbb{T}} \rightarrow \mathbf{Sub}_{\mathbf{Set}}^{\mathcal{K}}$ is the same data as a functor $\mathcal{K} \rightarrow \mathbf{Pt}(D_{\mathbb{T}})$. We write $M_k \in \mathbf{Pt}(D_{\mathbb{T}})$ for the image of $k \in \mathcal{K}$ and $f: M_k \rightarrow M_{k'}$ for the image of $f \in \mathcal{K}$.

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Lemma

The doctrine $\mathbf{Sub}_{\mathbf{Set}^{\mathcal{K}}}$ is a DL^+ -doctrine.

The complete join-primes of $\mathbf{Sub}_{\mathbf{Set}^{\mathcal{K}}}(X)$ are the *orbits* of elements of X , i.e. subpresheaves

$$\bar{x}(k') := \{x' \mid \exists f: k \rightarrow k'. X(f)(x) = x'\} \subseteq X(k'),$$

for some $x \in X(k)$.

Hence, the lifting $\alpha^{\sharp}: D_{\mathbb{T}}^{\delta} \Rightarrow \mathbf{Sub}_{\mathbf{Set}^{\mathcal{K}}} \circ M^{\mathrm{op}}$ exists.

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Being complete, each component $\alpha_X^{\sharp}$ has a left adjoint

$$\beta_X: \mathbf{Sub}_{\mathbf{Set}^{\mathcal{K}}}(MX) \rightarrow D_{\mathbb{T}}^{\delta}(X),$$

which moreover sends complete join-primes to complete join-primes.

Namely, given $m \in M_k^X$,

$$\beta_X(\bar{m}) = \text{tp}_{M_k}(m).$$

Strictly realising types and homogeneity

Proposition

For each X , $\alpha_X^\#$ is injective $\iff \beta_X$ is surjective,
 $\iff$ for each prime filter $\tau \subseteq D_{\mathbb{T}}(X)$, there exists
 $m \in M_k^X$ such that $\text{tp}_{M_k}(m) = \tau$.

We say that $\mathcal{K} \rightarrow \text{Pt}(D_{\mathbb{T}})$ *strictly realises all types*.

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Theorem (cf. Coumans)

The transformation $\alpha^\sharp: D_{\mathbb{T}}^\delta \Rightarrow \text{Sub}_{\text{Set}^{\mathcal{K}}} \circ M^{\text{op}}$ is a natural isomorphism if and only if $\mathcal{K} \rightarrow \text{Pt}(D_{\mathbb{T}})$ strictly realises all types and is homogeneous.

Countable, saturated models

As an application, $D_{\mathbb{T}}^{\delta}$ can be reconstructed, when $\mathbb{T}$ is ω -stable, from a particular subcategory of models of $\mathbb{T}$ equipped with ‘knowledge of the underlying sets’.

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Subcategories $\mathcal{A} \subseteq \mathbf{Mod}(\mathbb{T})$ and $\mathcal{B} \subseteq \mathbf{Mod}(\mathbb{T}')$
come equipped with forgetful functors

$$U: \mathcal{A} \subseteq \mathbf{Mod}(\mathbb{T}) \rightarrow \mathbf{Set}$$

and $U': \mathcal{B} \subseteq \mathbf{Mod}(\mathbb{T}) \rightarrow \mathbf{Set}.$

We write $\mathcal{A} \simeq_{\mathbf{Set}} \mathcal{B}$ if U and U' are equivalent
in $\mathbf{CAT}/\mathbf{Set}$.

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Similarly, there is a forgetful functor

$$\mathrm{Pt}(D_{\mathbb{T}}) \rightarrow \mathbf{Lex}[\mathbf{FinSet}^{\mathrm{op}}, \mathbf{Set}].$$

If $D_{\mathbb{T}} \cong D_{\mathbb{T}'}$, then

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Definition

A model $M \models \mathbb{T}$ is *countable and saturated* if M is countable and ω -saturated; let $\mathbf{Sat}_{\omega}(\mathbb{T}) \subseteq \mathbf{Mod}(\mathbb{T})$ denote the (full) subcategory of countable, saturated models.

Lemma

The category $\mathbf{Sat}_{\omega}(\mathbb{T})$ is always homogeneous.

A reconstruction result

Theorem (van Gool + W.)

Let $\mathbb{T}, \mathbb{T}'$ be such that every prime type is strictly realised in a countable, saturated model. Then,

$$D_{\mathbb{T}} \cong D_{\mathbb{T}'} \iff \mathbf{Sat}_{\omega}(\mathbb{T}) \simeq_{\mathbf{Set}} \mathbf{Sat}_{\omega}(\mathbb{T}').$$

($\Leftarrow$) If $\mathbf{Sat}_{\omega}(\mathbb{T}) \simeq_{\mathbf{Set}} \mathbf{Sat}_{\omega}(\mathbb{T}')$, then

$$\mathbf{Sub}_{\mathbf{Set}^{\mathbf{Sat}_{\omega}(\mathbb{T})}} \circ U \cong \mathbf{Sub}_{\mathbf{Set}^{\mathbf{Sat}_{\omega}(\mathbb{T}')}} \circ U'.$$

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Definition

A coherent theory $\mathbb{T}$ is ω -stable if, given any model $M \models \mathbb{T}$ and a countable subset $A \subseteq M$, there are countably many types with parameters in A .

Corollary

If $\mathbb{T}$ and $\mathbb{T}'$ are ω -stable, then prime types are strictly realised in countable, saturated models, and hence the reconstruction result applies.

Thank you for your attention

Our preprint: The points of canonical extensions of doctrines, [arXiv:2606.28736](#)

References and further reading

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